Sequences of measures in Bayesian inversion and their approximation

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Outline

Motivation: Filtering of model parameters

Bayesian filtering and smoothing
Applications

Sequential Monte Carlo Samplers

Other sequences

Continue reading

Motivation

Given a mathematical model $G : X \to H$

- H solution space
- X parameter space

G represents a process that is partially observed

- $\mathcal{O}_t : H \to Y_t$, observation operator at time t
- Y_t data space at time t
- $\mathcal{G}_t := \mathcal{O}_t \circ G : X \to Y_t$, forward response operator at time t

At time t, we observe $y_t = \mathcal{G}_t(\theta^\dagger) + \eta_t$

- $y_t \in Y_t$ is a data set
- $\eta_t \in Y_t$ is measurement noise
- $\theta^\dagger \in X$ is the true parameter

Problems

Smoothing or inverse problem

Identify $\theta^\dagger$ based on $\mathbf{y}_T := (y_t)_{t=1}^T$ using an 'estimator': $\theta(\mathbf{y}_T)$.

Filtering problem

Once new data is available, update the 'estimator'

$$(\theta(\mathbf{y}_T), y_{T+1}) \longrightarrow \theta(\mathbf{y}_{T+1})$$

Bayesian approach

Model $\theta \in X$ as a random variable

$\mathbb{P}(\theta \in \cdot) = \mu_0$ is called prior (measure)

the event $\{y_t = \mathcal{G}_t(\theta) + \eta_t\}$ occurs at time t

$\eta_t \sim \mu_{\text{noise}}^{(t)} := \mathrm{N}(0, \Gamma_t)$ follows a noise distribution

Tasks

Smoothing: Condition θ on the events $t = 1, ..., T$, i.e. compute

$$\mu_T := \mathbb{P}\left(\theta \in \cdot | \mathbf{y}_T\right) := \mathbb{P}\left(\theta \in \cdot \Big| \bigcap_{t=1}^{T} \{y_t = \mathcal{G}_t(\theta^\dagger) + \eta_t\}\right)$$

Filtering: Update $(\mu_T, y_{T+1}) \longrightarrow \mu_{T+1}$

μ_t is called posterior (measure) at time $t = 1, 2, 3, ...$

(Data) likelihood

By assumption, $y_t \sim \mathrm{N}(\mathcal{G}_t(\theta^\dagger), \Gamma_t)$.

The likelihood of observation y_t is given by

$$L_t(y_t|\theta) = \exp\left(-\frac{1}{2}\|\Gamma_t^{-1/2}\left(\mathcal{G}_t(\theta) - y\right)\|_2^2\right)$$

and the likelihood of $\mathbf{y}_T = (y_1, ..., y_T)$ is

$$\mathbf{L}_T(\mathbf{y}_T|\theta) = \prod_{t=1}^{T} L_t(y_t|\theta) = \exp\left(-\frac{1}{2}\sum_{t=1}^{T}\|\Gamma_t^{-1/2}\left(\mathcal{G}_t(\theta) - y\right)\|_2^2\right) \qquad (t = 1, ..., T)$$

Bayes' theorem

Bayes' theorem states

(Smoothing)
$$\frac{\mathrm{d}\mu_T}{\mathrm{d}\mu_0}(\theta) = \frac{\mathbf{L}_T(\mathbf{y}_T|\theta)}{\mu_0\left(\mathbf{L}_T(\mathbf{y}_T|\cdot)\right)}$$

(Filtering)
$$\frac{\mathrm{d}\mu_T}{\mathrm{d}\mu_{T-1}}(\theta) = \frac{L_T(y_T|\theta)}{\mu_{T-1}\left(L_T(y_T|\cdot)\right)},$$

$$\left[\quad \text{Notation:} \qquad \mu(f) = \int f \mathrm{d}\mu \quad\right]$$

Bayes' theorem

ce qui est le principe énoncé ci-dessus, lorsque toutes les causes sont *à priori* également possibles. Si cela n'est pas, en nommant p la probabilité *à priori* de la cause que nous venons de considérer; on aura $E = Hp$; et en suivant le raisonnement précédent, on trouvera

$$P = \frac{Hp}{S.Hp};$$

ce qui donne les probabilités des diverses causes, lorsqu'elles ne sont pas toutes, également possibles *à priori*.

Pour appliquer le principe précédent à un exemple, supposons qu'une urne renferme trois boules dont chacune ne puisse être que

Figure: Bayes' theorem in 'Théorie analytique des probabilités' by Pierre-Simon Laplace (1812, p. 182)

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Example: Elliptic inverse problem

Let $D = (0,1)^d$, $(d = 1,2,3)$. Define the model G by $C(D) \ni \theta \mapsto p \in H_0^1(D)$

(Darcy's law) $$-\exp(\theta(x))\nabla p(x) = \vec{q}(x) \quad (x \in D)$$

(Conservation) $$\nabla \cdot \vec{q}(x) = f(x) \quad (x \in D)$$

(Dirichlet BC) $$p(x) = 0 \qquad (x \in \partial D)$$

and the observation operator

(a bit sloppy) $$\mathcal{O}_t(p) = p(x_t), \qquad (t = 1, .., T)$$

G models the pressure head p in a groundwater reservoir D with log-permeability θ, source f, and $\vec{q}$ models the Darcy velocity

inverse problem: after T pressure measurements, estimate the log-permeability θ

smoothing problem: update the estimate after another measurement is taken

Example: Pendulum experiment

Consider a pendulum with known length 7.4, intial angle $\pi/36$, and initial speed 0.
Observations $y_t \in Y_t := \mathbb{R}$ are the angles at particular points in time $\tau_t \in [0, \infty)$
Estimate the gravitational acceleration $\theta^\dagger := g := 9.81$

Set-up:

Let $G : [0, 20] \to C[0, \infty)$ map $\theta \mapsto x$, where

$$\ddot{x}(t) = -\frac{\theta}{7.4}\sin(x(t)), \quad t \geq 0$$
$$\dot{x}(0) = 0$$
$$x(0) = \frac{\pi}{36}$$

Example: Pendulum experiment

Forward response operator: $\mathcal{G}_t(\theta) = x(\tau_t)$,

Measurement noise: $\eta_t \sim \mathrm{N}(0, 0.05^2)$,

Likelihood:

$$L_t(y_t|\theta) = \exp\left(-\frac{1}{2 \cdot 0.05^2}(x(\tau_t) - y_t)^2\right).$$

Prior is a truncated Gaussian:

$$\mu_0 := \frac{\mathrm{N}(10, 1^2)(\cdot \cap [0, 20])}{\mathrm{N}(10, 1^2)([0, 20])}$$

Example: Pendulum experiment

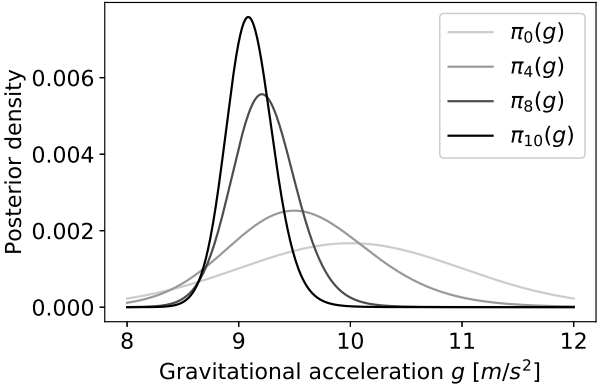

Figure: Prior density, posterior densities at $t = 4, 8, 10$

Example: Pendulum experiment

Latzer und Latzer

Wie lernen Computer die Erdbeschleunigung?

Computer lernen rätselhaft. Sie wissen, was Menschen im Internet verkaufen, Ihre erkennen unsere Gesichter, wenn wir das Smartphone entsperren, Sie merken sich unseren Tagesablauf und passen den Raumklima an. Der Mathematiker und Physiker Dr. Jonas Latz hat sich jetzt mit dem Phänomen auf der Plattform entsperrt und dort Wissenschaftler vorbei kurzer Zeit.

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Als Prof. Lin Weiss - Nach dem Seminar ist Dein Wissen Deine Kompetenz - Ich bin Dr. Jonas Latz. Ich bin Dozent an der Technischen Universität München und Mathematiker. Ich bin Physiker. Ich bin Ingenieur. Ich bin Informatiker. Ich bin ganz einfach ein Wissenschaftler.

Das Forschungsgebiet von Jonas Latz ist die Wahrscheinlichkeitstheorie. Seine Arbeit dreht sich um Stochastische Prozesse. Dabei geht es um Zufall und Unsicherheit. Latz hat zum Beispiel mit Kollegen aus Cambridge und Oxford ein Verfahren entwickelt, das Computer lernen lässt, wie sie Bilder von Gesichtern erkennen. Das Verfahren ist Teil der Forschungsgemeinschaft DFG, der Deutschen Forschungsgemeinschaft. Ziel ist es, Computer lernen zu lassen, wie sie Bilder von Gesichtern erkennen. Das Verfahren ist Teil der Forschungsgemeinschaft DFG, der Deutschen Forschungsgemeinschaft. Ziel ist es, Computer lernen zu lassen, wie sie Bilder von Gesichtern erkennen.

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Jonas Latz, Elisabeth Ullmann
www.tum.de/nachrichten/artikel/

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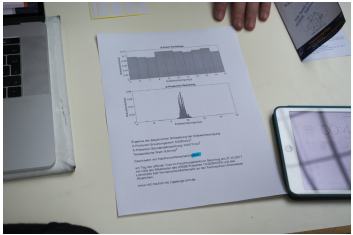
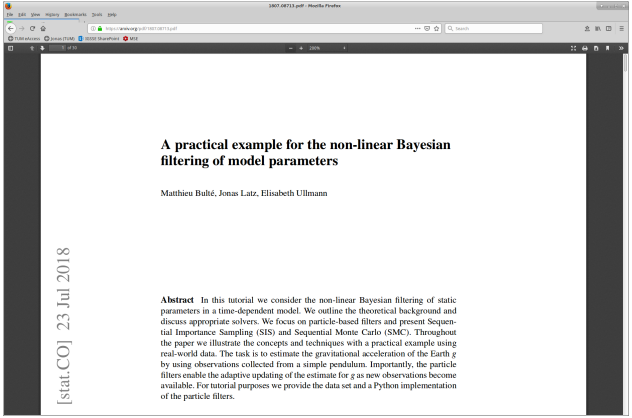


Open House Day Garching 2017, photos: Ionuţ-Gabriel Farcaş

Example: Pendulum experiment



[stat.CO] 23 Jul 2018

A practical example for the non-linear Bayesian filtering of model parameters

Matthieu Bulté, Jonas Latz, Elisabeth Ullmann

Abstract In this tutorial we consider the non-linear Bayesian filtering of static parameters in a time-dependent model. We outline the theoretical background and discuss appropriate solvers. We focus on particle-based filters and present Sequential Importance Sampling (SIS) and Sequential Monte Carlo (SMC). Throughout the paper we illustrate the concepts and techniques with a practical example using real-world data. The task is to estimate the gravitational acceleration of the Earth g by using observations collected from a simple pendulum. Importantly, the particle filters enable the adaptive updating of the estimate for g as new observations become available. For tutorial purposes we provide the data set and a Python implementation of the particle filters.

Bulté, L., Ullmann (2018): *A practical example for the non-linear Bayesian filtering of model parameters* Accepted in Springer Special Volume QUIET. Available in arXiv:1807.08713.

PYTHON-Code available in GITHUB/BayesianLearning/PenduSMC

Outline

Motivation: Filtering of model parameters

Sequential Monte Carlo Samplers

- Sequential Importance Sampling
- Sequential Monte Carlo
- SIS vs SMC

Other sequences

Continue reading

Summary: set-up[1]

Given a sequence of probability measures $(\mu_t)_{t=0}^T \in \mathrm{Prob}(X)^T$. Assume:

$\mu_t \ll \mu_{t-1}$ and Radon-Nikodym derivatives are

$$\frac{\mathrm{d}\mu_t}{\mathrm{d}\mu_{t-1}} \propto L_t(\theta) \qquad (t = 1, ..., T)$$

L_t stricly positive and bounded above $\quad (t = 1, ..., T)$
we can generate i.i.d. samples $\theta \sim \mu_0$

How to approximate $(\mu_t)_{t=0}^T$ by a sequence of discrete measures?

[Notation: $\mathrm{Prob}(X) := \{\mu | \mu$ is a probability measure on $(X, \mathcal{B}X)\}$]

[1] Del Moral, Doucet, Jasra 2006: *Sequential Monte Carlo samplers*, J. R. Statist. Soc. B 68(3), 411-436.

Weighting operator

Define the weighting operator

$$\mathrm{W}_t : \mathrm{Prob}(X) \to \mathrm{Prob}(X), \qquad \mu \mapsto \frac{\int_\bullet L_t(\theta)\mathrm{d}\mu(\theta)}{\int_X L_t(\theta)\mathrm{d}\mu(\theta)} \qquad (t = 1, ..., T)$$

This operator defines the update $\mu_t = \mathrm{W}_t(\mu_{t-1})$, $t = 1, ..., T$.

Sequential Importance Sampling (SIS)

Idea: Approximate μ_0 with Monte Carlo, then apply W_t iteratively

1: $t \leftarrow 1$,

2: Sample $(\theta^{(j)} : j = 1, ..., J) \sim \mu_0$ i.i.d. and set:

$$\widehat{\mu}_0 \leftarrow \frac{1}{J} \sum_{j=1}^{J} \delta(\cdot - \theta^{(j)}),$$

3: Update $\widehat{\mu}_t \leftarrow \mathrm{W}_t(\widehat{\mu}_{t-1})$

4: **If** $t < T$: $t \leftarrow t + 1$ and go to 3, **else:** terminate.

$$\left[\text{Notation:} \quad \text{Dirac measure} \quad \delta(A - x) = \begin{cases} 1, & \text{if } x \in A, \\ 0, & \text{otherwise.} \end{cases} \right]$$

Update discrete measure?

Let

$$\widehat{\mu}_{t-1} := \sum_{j=1}^{J} w_j \delta(\cdot - \theta^{(j)}).$$

Then,

$$\widehat{\mu}_t := \mathrm{W}_t(\widehat{\mu}_{t-1}) := \sum_{j=1}^{J} \left(\frac{L_t(\theta^{(j)}) \cdot w_j}{\sum_{k=1}^{J} L_t(\theta^{(k)}) \cdot w_k} \right) \delta(\cdot - \theta^{(j)})$$

Hence, when applying W_t, we actually update the weights:

$$w_j \leftarrow \frac{L_t(\theta^{(j)}) \cdot w_j}{\sum_{k=1}^{J} L_t(\theta^{(k)}) \cdot w_k}$$

(='classic importance sampling')

Error

We can quantify the error in SIS by

$$\mathrm{d}_{\mathrm{RMSE}}(\mu_t, \widehat{\mu}_t) := \sup_{|f|\leq 1} \sqrt{\mathbb{E}\left(\left|\mu_t(f) - \widehat{\mu}_t(f)\right|^2\right)} \leq 2 \cdot \sqrt{\frac{\mu_0(\mathbf{L}_t^2)}{J \cdot \mu_0(\mathbf{L}_t)^2}} = 2 \cdot \sqrt{\frac{1}{\mathrm{ESS}_t}}$$

$\mathrm{d}_{\mathrm{RMSE}}$ computes the root mean square error between integrals over $\mu_t, \widehat{\mu}_t$

it is necessary since $\widehat{\mu}_t$ is a random probability measure

$\mathrm{ESS}_t := \frac{J \cdot \mu_0(\mathbf{L}_t)^2}{\mu_0(\mathbf{L}_t^2)}$ is called effective sample size

Problem: in many examples $\mathrm{ESS}_t \to 0$, as $t \to \infty$.

Example: degeneracy over time

Consider the filtering problem given prior measure

$$\mu_0 := \mathrm{N}(0, 1^2)$$

and likelihoods

$$L_t(y_t|\theta) = \exp\left(-\frac{1}{2}\|\theta - y_t\|_2^2\right).$$

(given samples $(y_1, ..., y_T)$ from a Gaussian measure $\mathrm{N}(\theta^\dagger, 1^2)$, find $\theta^\dagger$.)

Then,

$$\mathrm{ESS}_t = O(J \cdot t^{-1/2}; t \to \infty).$$

degenerates eventually independently of fixed J.

Error

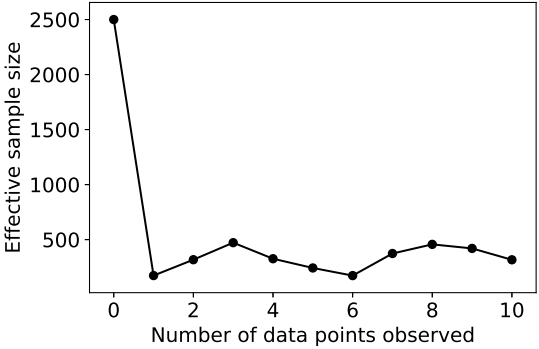


Figure: ESS over time in a SIS approximation of the pendulum filtering problem, with $J = 2500$ samples

Problem: $\mu_0 \not\approx \mu_t$, if t large, but we approximate μ_t with samples from μ_0

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Sequential Monte Carlo (SMC)

Target: Retain large effective sample size

Techniques:

Move particles with MCMC moves
Remove particles with small weights

Sequential Monte Carlo: A genetic algorithm?!

Let $\widehat{\mu}_0$ be given as in SIS.

For $t = 1, ..., T$:

(Learning) Apply weighting operator W_t

(Selection) Remove particles with small weights by resampling

(Mutation) Move particles with MCMC

Selection

We approximate

$$\widehat{\mathrm{ESS}}_t = \frac{\left(\sum_{j=1}^{J} w_j\right)^2}{\left(\sum_{j=1}^{J} w_j^2\right)} \approx \frac{J \cdot \mu_0(\mathbf{L}_t)^2}{\mu_0(\mathbf{L}_t^2)} = \mathrm{ESS}_t$$

If $\widehat{\mathrm{ESS}}_t \approx J$, the particles approximate μ_t well.

Else if $\widehat{\mathrm{ESS}}_t \not\approx J$, resample

$$\widehat{\theta}^{(j)} \sim \widehat{\mu}_t, \quad j = 1, \ldots, J \quad \text{and set } \widehat{\mu}_t \leftarrow \sum_{j=1}^{J} \frac{1}{J} \delta(\cdot - \widehat{\theta}^{(j)})$$

We denote this procedure by $S : \mathrm{Prob}(X) \times \mathcal{B}\mathrm{Prob}(X) \to [0, 1]$

Mutation

Let

$$K_t : X \times \mathcal{B}X \to [0,1]$$

be a Markov kernel from $(X, \mathcal{B}X)$ into itself.

we assume stationarity of K_t with respect to μ_t, i.e.

$$\mu_t K_t = \int K_t(\theta, \cdot) \mathrm{d}\mu_t(\theta) = \mu_t$$

like an MCMC kernel (Gibbs, Metropolis-Hastings, HMC,...)

We apply K_t to all particles and obtain

$$\widehat{\theta}^{(j)} \sim \delta(\cdot - \theta^{(j)}) K_t, (j = 1, ..., J)$$

$$\widehat{\mu}_t \leftarrow \sum_{j=1}^{J} w_j \delta(\cdot - \widehat{\theta}^{(j)})$$

We denote this step by $M_t : \mathrm{Prob}(X) \times \mathcal{B}\mathrm{Prob}(X) \to [0,1]$

Sequential Monte Carlo

1: $t \leftarrow 1$,

2: Sample $(\theta^{(j)} : j = 1, ..., J) \sim \mu_0$ i.i.d. and set:

$$\widehat{\mu}_0 \leftarrow \frac{1}{J} \sum_{j=1}^{J} \delta(\cdot - \theta^{(j)}),$$

3: Learn $\widehat{\mu}_t \leftarrow \mathrm{W}_t(\widehat{\mu}_{t-1})$

4: Select $\widehat{\mu}_t \sim \mathrm{S}(\widehat{\mu}_t, \cdot)$

5: Mutate $\widehat{\mu}_t \sim \mathrm{M}_t(\widehat{\mu}_t, \cdot)$

6: **If** $t < T$: $t \leftarrow t + 1$ and go to 3, **else:** terminate.

Error

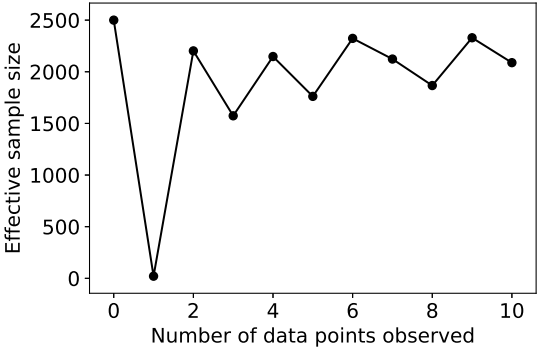

Figure: ESS over time in an SMC approximation of the pendulum filtering problem, with $J = 2500$ samples, and resampling if $\widehat{\mathrm{ESS}}_t < 0.75 \cdot J$

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Estimation of posterior mean

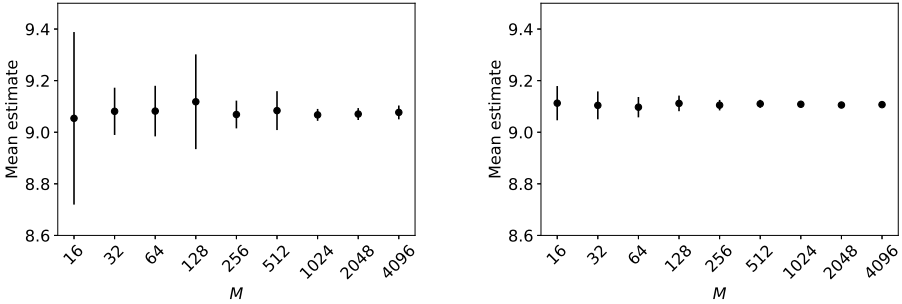
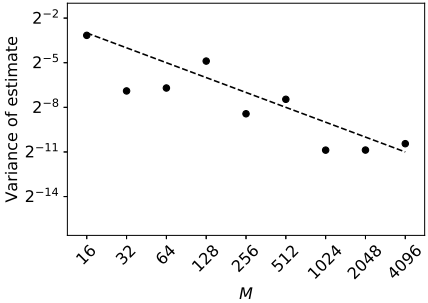


Figure: Sample mean and sample standard deviation of posterior mean estimates through 50 runs in the pendulum example. Left: SIS, Right: SMC.

Variance posterior mean estimator



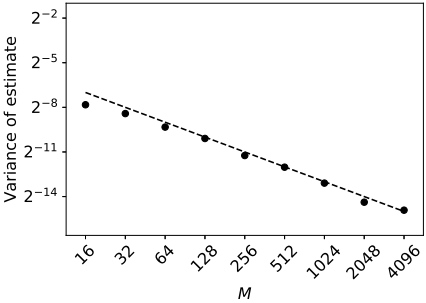


Figure: Variance of posterior mean estimates through 50 runs in the pendulum example. Left: SIS, Right: SMC.

Outline

Motivation: Filtering of model parameters

Sequential Monte Carlo Samplers

Other sequences

- Bayesian statistics
- Estimating rare events

Continue reading

Other sequences

Sequential Monte Carlo algorithm is efficient in filtering situations

Can we construct sequences artificially to profit from the efficient SMC algorithm?

Problem setting

Consider a Bayesian inverse problem with a single data set y.

Given prior $\mu_0 \in \mathrm{Prob}(X)$ and likelihood $L(y|\theta)$

Compute posterior $\mu^y \in \mathrm{Prob}(X)$:

$$\frac{\mathrm{d}\mu^y}{\mathrm{d}\mu_0}(\theta) \propto L(y|\theta)$$

Standard idea: Apply importance sampling (i.e. compute $\mathrm{W}_1(\widehat{\mu}_0)$)

Importance Sampling

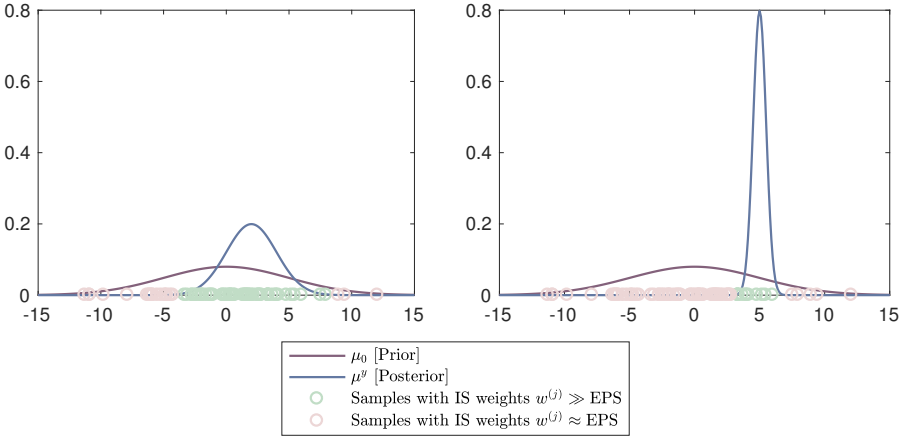


Figure: Importance sampling with a flat posterior (left) and with a concentrated posterior (right)

Tempering

Construct a sequence $\mu_0, ..., \mu_T$,

μ_0 is the prior distribution

$\mu_T = \mu^y$ is the posterior distribution

μ_t is accessible by importance sampling from μ_{t-1}.

Use a tempering[2] of the likelihood ($\beta_t \in [0, 1], \beta_0 = 0, \beta_T = 1$):

$$\frac{\mathrm{d}\mu_t}{\mathrm{d}\mu_0}(\theta) \propto L(y|\theta)^{\beta_t}.$$

$(\beta_t : t = 1, ..., T)$ can be determined a priori or adaptively

[2] Neal 2001: *Annealed Importance Sampling*, Stat. Comp. 11(2), 125-139.

Tempering

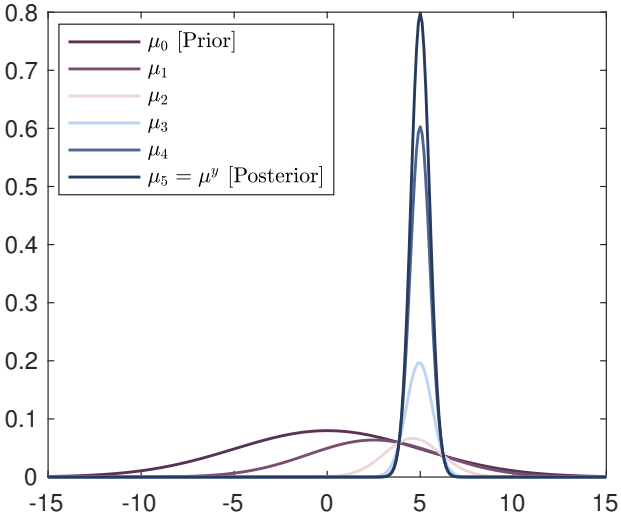


Figure: Intermediate probability measures.

Tempering

requires few model evaluations (in practice)

models can be evaluated in parallel

enables Bayesian inversion with computationally expensive forward models:

- Navier-Stokes equation[3]
- Elliptic equation in 3D[4]
- Cahn-Hilliard equation for tumour growth[5]

[3] Kantas, Beskos, Jasra: *Sequential Monte Carlo Methods for High-Dimensional Inverse Problems: A case study for the Navier-Stokes equations*, SIAM/ASA J. Uncertain. Quantif. 2(1), 464-489.

[4] Beskos, Jasra, Muzaffer, Stuart 2015: *Sequential Monte Carlo methods for Bayesian elliptic inverse problem*, Stat. Comput. 25(4), 727-737.

[5] Kahle, Lam, L., Ullmann 2018: *Bayesian parameter identification in Cahn–Hilliard models for biological growth*, under review.

(Multilevel) Bridging

In realistic problems: Approximate $L_\ell \approx L$

$\ell \in \{1, ..., N_{\mathrm{L}}\}$ reflects the accuracy of the underlying mathematical model

Task: Increase the accuracy of μ_ℓ^y, update: $\mu_\ell^y \longrightarrow \mu_{\ell+1}^y$.

Bridging sampler[6]

$$\frac{\mathrm{d}\mu_t}{\mathrm{d}\mu_0}(\theta) \propto L_\ell(y|\theta)^{1-\zeta_t} L_{\ell+1}(y|\theta)^{\zeta_t},$$

where μ_0 is the underlying prior distribution.

$(\zeta_t : t = 1, ..., T)$ can be determined a priori or adaptively

[6] Koutsourelakis 2009: *A multi-resolution, non-parametric, Bayesian framework for identification of spatially-varying model parameters*, J. Comp. Phys. 228(17), 6184-6211.

(Multilevel) Bridging

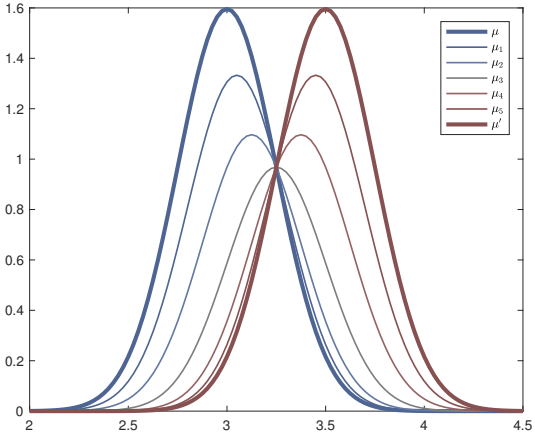


Figure: Bridging between two probability measures μ and μ'

However...

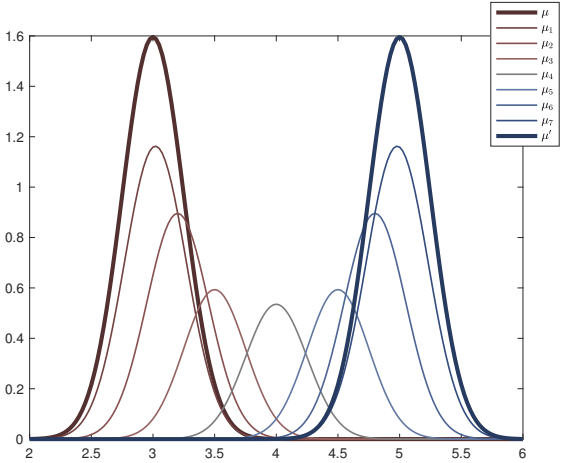


Figure: Bridging between two distributions μ and μ' with a high discrepancy

Multilevel Sequential^2 Monte Carlo

Bridging is often not very efficient

μ_ℓ^y is very concentrated and biased
high computational cost and low accuracy in $\mu_\ell^y \longrightarrow \mu_{\ell+1}^y$

Instead: Combine Bridging and Tempering

Multilevel Sequential^2 Monte Carlo sampler[7]:

$$\frac{\mathrm{d}\mu_t}{\mathrm{d}\mu_0}(\theta) \propto (L_\ell(y|\theta)^{1-\zeta_t} L_{\ell+1}(y|\theta)^{\zeta_t})^{\beta_t},$$

where μ_0 is the underlying prior distribution.

$((\beta_t, \zeta_t) : t = 1, ..., T)$ can be determined a priori or adaptively

[7] L., Papaioannou, Ullmann 2018: *Multilevel Sequential^2 Monte Carlo for Bayesian Inverse Problems*, J. Comput. Phys. 368, 154-178.

Outline

Motivation: Filtering of model parameters

Sequential Monte Carlo Samplers

Other sequences

- Bayesian statistics
- Estimating rare events

Continue reading

Rare events

Let $\theta \sim \mu_0$ and $f : X \to \mathbb{R}$. $\{F(\theta) \leq 0\}$ is an event of failure

Estimate

$$p_f = \mathbb{P}(F(\theta) \leq 0),$$

which is a very small number $p_f \approx 10^{-7}, 10^{-8}, 10^{-9}$.

Rare events

Standard Monte Carlo approach: Draw J i.i.d. samples $(\theta^{(j)})_{j=1}^{J}$ from μ_0 and estimate

$$\widehat{p}_f = \frac{\#\{j : F(\theta^{(j)}) \leq 0\}}{J}$$

However, Chebyshev inequality $\Rightarrow$

$$\mathbb{E}[(\widehat{p}_f - p_f)^2] \leq \frac{p_f(1-p_f)}{J}$$

$$\Rightarrow \mathbb{E}[(\widehat{p}_f - p_f)^2]/p_f^2 \leq \frac{1}{p_f \cdot J}$$

Example: If $p_f = 10^{-8}$, we need $J = 10^9$ samples for a relative MSE of 10%. (Too many.)

Subset simulation[8]

Given a strictly decreasing sequence $B_1 > \cdots > B_T = 0$ (typically defined adaptively)

Use SMC to approximate

$$\mu_t = \mathbb{P}(\theta \in \cdot | F(\theta) \leq B_t), \qquad \frac{\mathrm{d}\mu_t}{\mathrm{d}\mu_0}(\theta) \propto \begin{cases} 1, & \text{if } F(\theta) \leq B_t, \\ 0, & \text{otherwise.} \end{cases}$$

By-products (remember normalising constants?):

$$p^{(t)} = \mathbb{P}(F(\theta) \leq B_{t+1} | F(\theta) \leq B_t) = \mu_t(F(\theta) \leq B_{t+1}) \qquad t \geq 1$$
$$p^{(0)} = \mathbb{P}(F(\theta) \leq B_1) = \mu_0(F(\theta) \leq B_1)$$

Estimate: $\widehat{p}_f^{\mathrm{SuS}} = \prod_{t=0}^{T} p^{(t)}$

[8] Au, Beck 2001: *Estimation of small failure probabilities in high dimensions by subset simulation*, Probabilistic Engineering Mechanics 16, 263-277.

Subset simulation

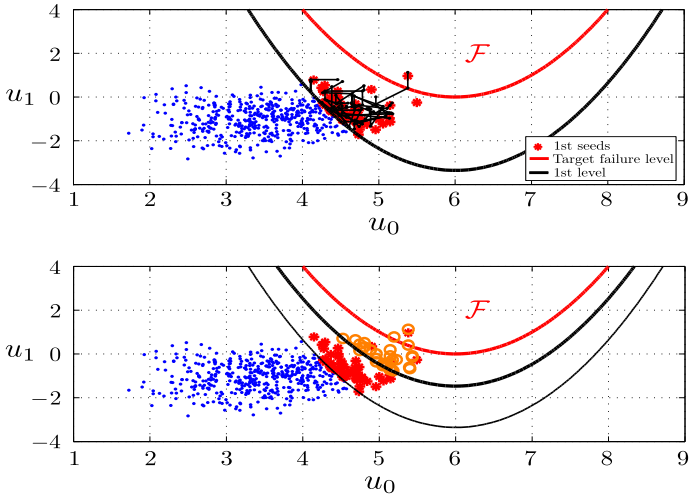


Figure: Subset simulation (from Felipe Uribe)

Subset simulation

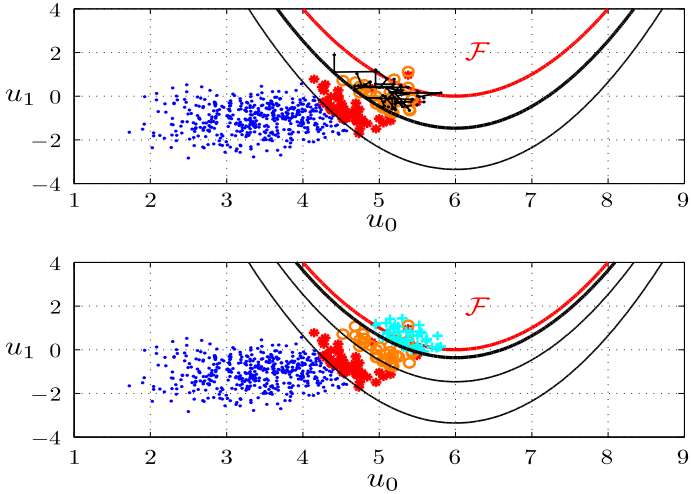


Figure: Subset simulation (from Felipe Uribe)

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Papers cited on these slides: `http://www.latz.io`

Webpage of Prof. Arnaud Doucet (Oxford, UK):
`http://www.stats.ox.ac.uk/~doucet/smc_resources.html`

Webpage of Prof. Pierre Del Moral (Bordeaux, France):
`http://people.bordeaux.inria.fr/pierre.delmoral/simulinks.html`

Jonas Latz

`Input/Output: www.latz.io`